

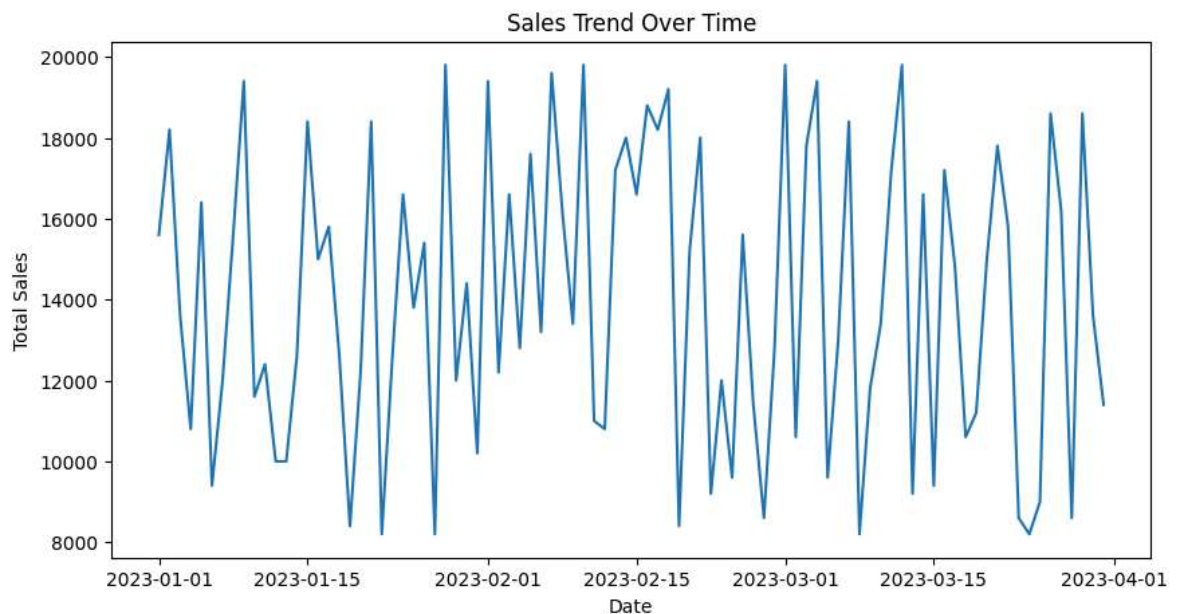
```
In [2]: import pandas as pd                #Step 1:Import Libraries
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error,r2_score
```

```
In [11]: df=pd.read_csv('C:/Users/hp/Desktop/sales_data.csv') #Step 2:Load Dataset
print(df.head())
```

	Date	Product	Units Sold	Unit Price	Total Sales
0	2023-01-01	Product A	78	200	15600
1	2023-01-02	Product A	91	200	18200
2	2023-01-03	Product A	68	200	13600
3	2023-01-04	Product A	54	200	10800
4	2023-01-05	Product A	82	200	16400

```
In [13]: df['Date']=pd.to_datetime(df['Date']) #Step 3:Convert date to Numeric for Model
df['Day']=df['Date'].dt.dayofyear
```

```
In [14]: plt.figure(figsize=(10,5))          #Step 4:Plot the Sales Trend:this helps to vi
sns.lineplot(data=df,x='Date',y='Total Sales')
plt.title('Sales Trend Over Time')
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.show()
```



```
In [15]: #Step 5:Split Data into Feautres & Target:
x=df[['Day']]          #Feautre(day of year)
y=df['Total Sales']    #Target(sales amount)
```

```
In [16]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
In [17]: model=LinearRegression()          #Step 7:Train Linear Regression Model
model.fit(x_train,y_train)
```

Out[17]:

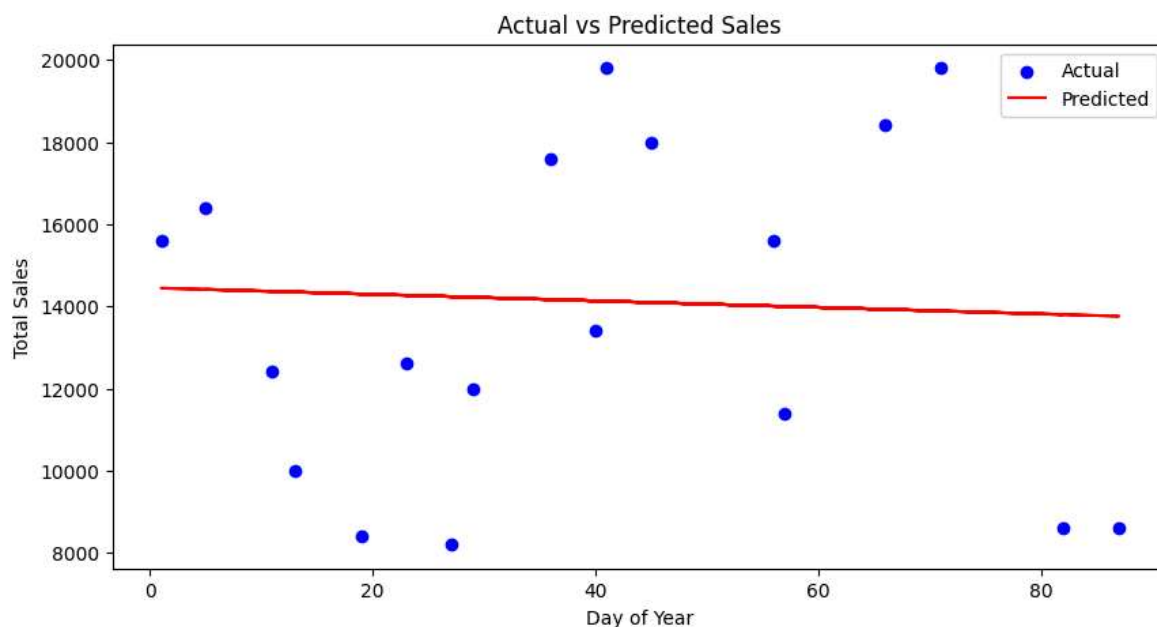
▼ LinearRegression ⓘ ?

LinearRegression()

In [18]: `y_pred=model.predict(x_test)` *#Step 8:Predict Sales on Test data*In [20]: `print('Mean Squared error:',mean_squared_error(y_test,y_pred))` *#Step 9:Evaulat*
`print('R2 Score:', r2_score(y_test, y_pred))`

Mean Squared error: 15819316.567337014

R2 Score: -0.01718210551098509

In [22]: `plt.figure(figsize=(10,5))` *#Step 10:Plot Actual vs Predicted Sales*
`plt.scatter(x_test, y_test, color='blue', label='Actual')`
`plt.plot(x_test, y_pred, color='red', label='Predicted')`
`plt.title("Actual vs Predicted Sales")`
`plt.xlabel("Day of Year")`
`plt.ylabel("Total Sales")`
`plt.legend()`
`plt.show()`In [23]: `future_day = pd.DataFrame({'Day': [100]})`
`future_prediction = model.predict(future_day)`
`print("Predicted Sales on Day 100:", future_prediction[0])`

Predicted Sales on Day 100: 13656.903970867063

In []: